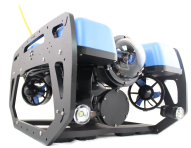
Calibrate BlueROV2 from measured motion

(a) Fit the dynamics

5 calibration runs

Same PWM input

6 recorded channels



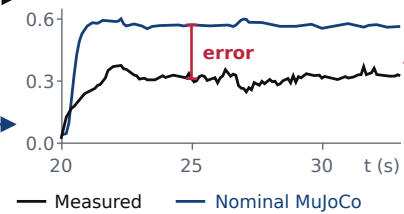
BlueROV2 Standard

T200 map + logged voltage

6-DOF model · 50 Hz

Compare responses

Surge example (m/s)



Compute the update

Weighted residual r

Scale and stack errors

Jacobian J

Finite differences

Solve for Δθ

$r(\theta + \Delta\theta) \approx r(\theta) + J\Delta\theta$

Bounded least squares

Update 24 parameters θ ← θ + Δθ

Drag, buoyancy and thruster gains

(b) Test on new runs

4 held-out runs

Select parameters on full calibration replays, then freeze them.

Held-out PWM → **MuJoCo replay · 1 ms** → **Compare with measured motion** → **6-axis RMSE**

Product photo: Blue Robotics Media Kit.